

HD189733b

R-0015

Benchmark run · workflow W8-benchmark · record 2026-bench_HD189733_260704 · created 2026-07-08T06:31:05+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461225.7926 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.187 +/- 0.055
Transit depth (Rp/Rs)^2	3.48 +/- 2.06 [%]
Orbital Inclination (inc)	87.19 +/- 2.59
Airmass coefficient 1 (a1)	0.34 +/- 0.17
Airmass coefficient 2 (a2)	1.04 +/- 0.65
Scatter in the residuals of the lightcurve fit is	51.84 %
Best Comparison Star	None
Optimal Aperture	2.76
Optimal Annulus	3.0
Transit Duration (day)	0.078 +/- 0.021

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.187 ± 0.055 vs 0.15534 (deviation 0.44σ)
Duration fit vs published	0.078 d vs 0.0754 d (deviation 0.09σ)
Mid-time O–C	-4.3 min
Depth SNR	2.6
Residual scatter	51.84 %

Light curve

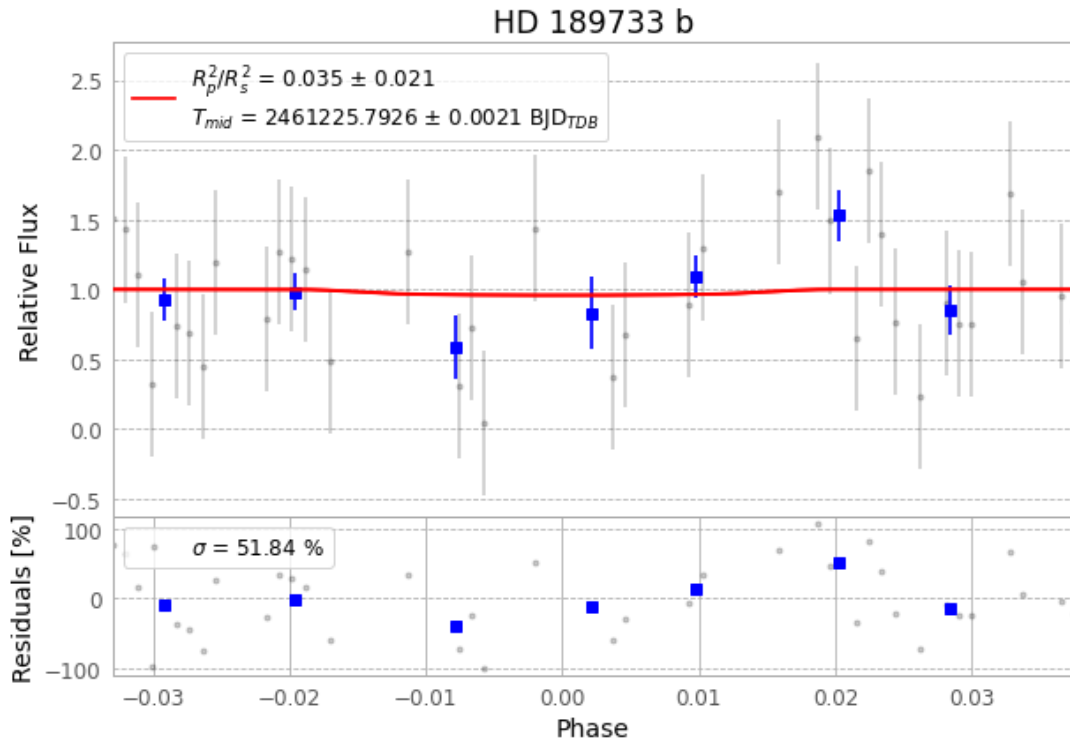


Figure 1 — FinalLightCurve_HD 189733 b_2026-07-04.png (SHA-256 8450a1f5cf6f9581...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [348, 408], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 c25c7ef9eeb2e969...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

78 FITS frames (51 MB), HD189733260704050417.FITS → HD189733260704085416.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HD189733-260704.log (8e2de097c8e9...), FinalLightCurve_HD 189733 b_2026-07-04.png (8450a1f5cf6f...)

Compute resources

Wall time 195.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-15 16:40Z.